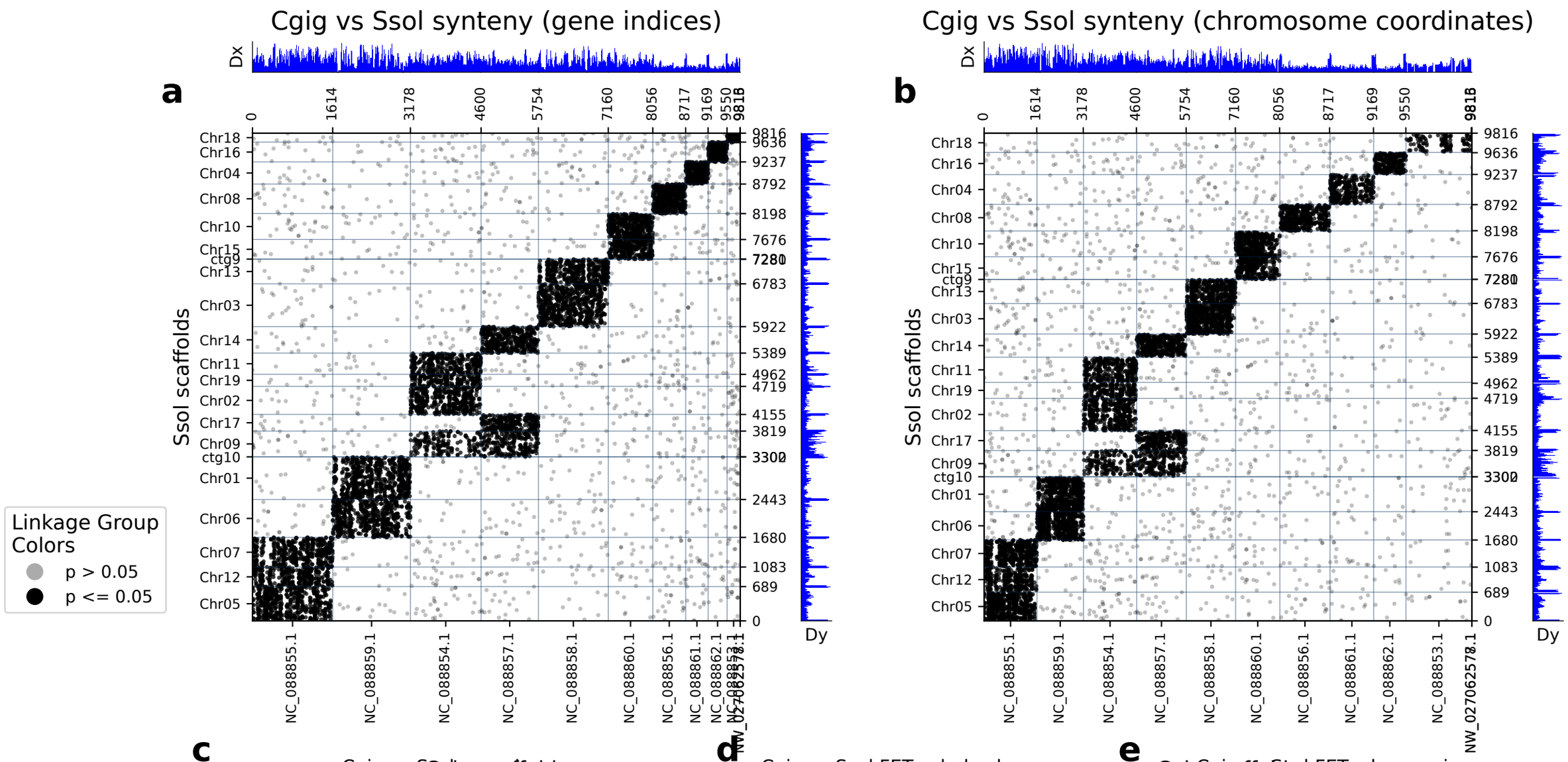
diamond: 9817 orthologs on 12 Cgig scaffolds and 21 Ssol scaffolds.



c

Cgig vs Ssol FET, chrom. pieces

d

Cgig vs Ssol FET, whole chroms.

	Cgig scaf	Ssol scaf	p (FET)	Count
0	NC_088858.1	Chr03	0.00e+00	768
1	NC_088859.1	Chr01	0.00e+00	755
2	NC_088859.1	Chr06	0.00e+00	665
3	NC_088855.1	Chr05	0.00e+00	582
4	NC_088856.1	Chr08	0.00e+00	501
5	NC_088855.1	Chr07	1.66e-295	488
6	NC_088854.1	Chr02	0.00e+00	486
7	NC_088857.1	Chr14	0.00e+00	458
8	NC_088858.1	Chr13	0.00e+00	446
9	NC_088860.1	Chr10	0.00e+00	431
10	NC_088854.1	Chr11	6.86e-237	356
11	NC_088861.1	Chr04	0.00e+00	345
12	NC_088860.1	Chr15	4.53e-288	326
13	NC_088855.1	Chr12	2.24e-186	321
14	NC_088862.1	Chr16	0.00e+00	309
15	NC_088857.1	Chr17	4.13e-209	279
16	NC_088857.1	Chr09	6.91e-96	246
17	NC_088854.1	Chr19	2.59e-117	194
18	NC_088854.1	Chr09	1.54e-30	180
19	NC_088853.1	Chr18	6.88e-157	122

e

Cgig vs Ssol FET, chrom. pieces

	Cgig scaf	Ssol scaf	p (FET)	Count
0	NC_088858.1 : 0-end	Chr03 : 0-end	0.00e+00	768
1	NC_088859.1 : 0-end	Chr01 : 0-end	0.00e+00	755
2	NC_088859.1 : 0-end	Chr06 : 0-end	0.00e+00	665
3	NC_088855.1 : 0-end	Chr05 : 0-end	0.00e+00	582
4	NC_088856.1 : 0-end	Chr08 : 0-end	0.00e+00	501
5	NC_088855.1 : 0-end	Chr07 : 0-end	1.66e-295	488
6	NC_088854.1 : 0-end	Chr02 : 0-end	0.00e+00	486
7	NC_088857.1 : 0-end	Chr14 : 0-end	0.00e+00	458
8	NC_088858.1 : 0-end	Chr13 : 0-end	0.00e+00	446
9	NC_088860.1 : 0-end	Chr10 : 0-end	0.00e+00	431
10	NC_088854.1 : 0-end	Chr11 : 0-end	6.86e-237	356
11	NC_088861.1 : 0-end	Chr04 : 0-end	0.00e+00	345
12	NC_088860.1 : 0-end	Chr15 : 0-end	4.53e-288	326
13	NC_088855.1 : 0-end	Chr12 : 0-end	2.24e-186	321
14	NC_088862.1 : 0-end	Chr16 : 0-end	0.00e+00	309
15	NC_088857.1 : 0-end	Chr17 : 0-end	4.13e-209	279
16	NC_088857.1 : 0-end	Chr09 : 0-end	6.91e-96	246
17	NC_088854.1 : 0-end	Chr19 : 0-end	2.59e-117	194
18	NC_088854.1 : 0-end	Chr09 : 0-end	1.54e-30	180
19	NC_088853.1 : 0-end	Chr18 : 0-end	6.88e-157	122

(a) depicts the chromosomes/scaffolds of Cgig (x-axis) plotted against the chromosomes/scaffolds of Ssol (y_axis). Each dot in the plot represents an ortholog, specifically a reciprocal best diamond blastp match between two species. The unit of the x- and y-axes are the number of orthologous proteins between these two species: 9817 orthologs found between 12 Cgig scaffolds and 21 Ssol scaffolds. If there are chromosome breaks, Fisher's exact test is used to calculate the significance of the interactions between the sub-chromosomal pieces. Otherwise Fisher's exact test is calculated on whole chromosomes. The opacity of the dots depict the significance from Fisher's exact test. Dots that are a solid color are in cells with a FET p-value less than or equal to 0.05. Dots that are translucent are in cells with a FET p-value greater than 0.05. The Dx and Dy values depict places where there may be sudden breaks in synteny. See the supplementary information the following paper for more information on Dx and Dy: Simakov, Oleg, et al. Deeply conserved synteny resolves early events in vertebrate evolution. Nature Ecology & Evolution 4.6 (2020): 820-830. (b) depicts the same information as panel a, but it is plotted in the organisms chromosome basepair coordinates rather than gene index. This is useful for visualizing gene-poor regions of the chromosomes (c) shows which gene groups are most prevalent in each chromosome pair. The FET p-value in this table corresponds to the whole-chormosome FET p-value , and is not a FET value of the correlation between the chromosome pair and the gene group. (d) shows chromosome-scale significance values. This table shows the same information as c, but does not factor in gene group information. (e) shows the FET p-values of the sub-chromosomal compartments. Useful for showing if single arms of chromosomes are correlated with other regions.